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SUPERSEDING
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FEDERAL SPECIFICATION

DETERGENT, LABORATORY GLASSWARE AND SURGICAL INSTRUMENTS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers powdered or granular detergents for use in cleaning laboratory glassware and surgical instruments (see 6.1).

1.2 Classification. The detergent covered by this specification shall be of one type only.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-S-766 - Steel Plates, Sheets, and Strip-Corrosion Resisting.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act.
40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials:

D 502 - Particle Size of Soaps and Other Detergents.
D 515 - Phosphorus in Water.
D 820 - Chemical Analysis of Soaps Containing Synthetic Detergents.
D 1331 - Surface and Interfacial Tension of Solutions of Surface-Active Agents.
E 70 - pH of Aqueous Solutions with the Glass Electrode.

3. REQUIREMENTS

3.1 Material. The detergent covered by this specification shall be granular in form, and free flowing without caking. The detergents shall be formulated to remove biological and chemical soils from glassware and stainless steel. The detergent shall be completely soluble in water.

3.2 Basic requirements. The detergent shall conform to the requirements shown in table I when tested as specified in table II.

TABLE I. Basic requirements

Requirement	Value	
Corrosion 1/	2.0	mg (max.)
Percent retained on No. 20 sieve	1	percent (max.)
Percent retained on No. 100 sieve	70	percent (min.)
pH	7.0	(min.)
	11.0	(max.)
Interfacial tension	40	dynes/cm (max.)
Polyphosphates	4	percent (min.)

1/ The detergents shall not cause discoloration, etching, or a film deposit on the test strip as a result of the performance of the test.

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3.3 Cleaning efficiency. When tested as specified in 4.4.4, the cleaner shall clean a minimum of four out of five of the test beakers. Failure is determined when any visible streaks, residue or soil remains after cleaning.

3.4 Rinsibility. The detergent shall not leave any detectable residue, or allow water droplet formation when tested as specified in 4.4.9.

3.5 Solubility. The detergent shall demonstrate complete solubility in hard water when tested as specified in 4.4.10.

3.6 Water softening. A 0.25 percent solution of the detergent shall be able to completely soften 30-grain hard water when tested as specified in 4.4.11.

3.7 Labeling. The product label shall be required to include the instructions for normal use dilution and the appropriate caution statements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection sampling. Sampling for the inspection procedures set forth in this specification shall be performed in accordance with MIL-STD-105. The lots to be inspected shall consist of no more than one classification (see 1.2) and one package size (see 5.1).

4.3 Examination.

4.3.1 Examination of preparation for delivery. The sample unit shall be defined as one shipping container fully marked and prepared for delivery. The samples shall be selected in accordance with inspection level I and the AQL shall be 4.0 percent defective.

4.3.1.2 Examination procedure. The examination shall be made to determine the compliance of the packing and marking with the requirements of 5.2 and 5.3. The sample units shall be examined for the defects described below and scored accordingly.

Examination of preparation for delivery

Examine	Defects
Packing	Not as specified (see 5.2).
Marking	Omitted, incorrect; illegible; improper size, location, sequence or method of application)see 5.3).
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of packing container or its component materials.

4.3.2 Examination of contents of the unit container.

4.3.2.1 Sampling plan. The sample unit shall be defined as one such filled container. The sample units shall be selected in accordance with level S-3 and the AQL shall be 4.0 percent defective.

4.3.2.2 Examination procedure. The examination shall be made to determine the compliance of the packaging and contents with the requirements of 5.1, 5.3, and 3.1. The sample units shall be examined for the defects described below and scored accordingly.

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Examination of the unit container

Examine	Defect
Packaging	Not as specified (see 5.1).
Marking	Omitted, incorrect; illegible; improper size, location, sequence or method of application (see 3.7 and 5.3).
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers.
Fill	Incomplete weight (see 5.1).
Contents	Not free flowing; not homogeneous (see 3.1).

4.4 Testing.

4.4.1 Sampling for testing. The sample unit shall be defined as one such filled container. A one pound composite shall be made by mixing equal portions of each of the sample units taken from the lot in accordance with inspection level S-1. The lot shall be unacceptable if the composite sample fails to pass any of the tests described in 4.4.2.

4.4.2 Testing procedures. The composite sample shall be tested in accordance with the instructions in table II.

TABLE II. Instructions for testing the detergent

Characteristic	Spec. reference: Require-ment	Spec. reference: Test Method	Results reported as: Pass or Fail	Results reported as: Numerically to nearest
Corrosion 1/	Table I	4.4.3		0.1 mg
Cleaning efficiency	Table I	4.4.4	X	
Percent retained on No. 20 sieve	Table I	4.4.5		0.1 percent
Percent retained on No. 100 sieve	Table I	4.4.5		0.1 percent
pH	Table I	4.4.6		0.1
Interfacial tension	Table I	4.4.7		0.1 dyne
Polyphosphates	Table I	4.4.8		0.1 percent
Rinsibility	3.4	4.4.9	X	
Solubility	3.5	4.4.10	X	
Water softening	3.6	4.4.11	X	

1/ Examine the test strip at the conclusion of the test for evidence of discoloration, etching, or a film deposit.

4.4.3 Corrosion. Polish by buffing two strips, 1 cm wide by 8 cm long, of each of the following corrosion-resisting steels, classes 302, 304, and 430, condition A, finish No. 4, conforming to QQ-S-766. Degrease the strips by swabbing in a mixture of equal parts of alcohol and acetone. Allow the solvent to evaporate, heat in an oven at 105° + 2°C for 30 minutes, cool in a desiccator. Remove the strips from the desiccator and quickly weigh them to the nearest 0.1 mg. Into test tubes measuring 32 mm by 300 mm, pour 160 ml of a mixture of 6.5 g of the compound and 250 ml of distilled water. Completely immerse the sample strips in the test solutions and maintain them at 175°F for a period of 2 hours. At the end of 2 hours, remove the strips, rinse thoroughly with tap water, then with acetone, and dry for 30 minutes in an oven at 105° + 2°C. Cool in a desiccator to room temperature. Remove the strips from the desiccator and quickly reweigh them to the nearest 0.1 mg and examine for evidence of corrosion such as pitting, etching, or discoloration. Calculate the change of weight of strips as milligrams per square centimeter per hour of exposed surface.

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4.4.4 Cleaning efficiency. Prepare 500 ml of a 1 percent solution of the test cleaning compound in hard water containing 0.264 gm/liter $CaCl_2.2H_2O$ and 0.295 gm/liter $MgSO_4.7H_2O$. Using a Waring blender, emulsify 40 gm fresh beef liver, 20 gm fresh eggwhite, 10 gm of hydrogenated vegetable oil, and 80 ml of distilled water. Pour the emulsified soil into five clean 250 ml beakers and pour out the soil while coating the entire interior of each beaker as uniformly as possible. In order to soil the beakers efficiently, pour all the soil from one beaker to the next, until all five are soiled. The test beakers shall be sufficiently clean before soiling so that when rinsed with water, beads of water shall not form on the surface. Support each beaker in an inverted position on two glass stirring rods and allow to dry overnight. Set each beaker upright and pour 100 ml of the test cleaner (at 20° + 2°C) into each one without allowing the liquid to touch the sides above the 100 ml mark. Let stand 30 minutes at 20° + 2°C. Decant the solution and rinse twice with 100 ml portions of distilled water and allow to dry in an inverted position. Examine the bottom of each beaker for streaks, or soiled areas. Any visible streaks, residue or soil shall be cause for failure. If more than one beaker fails, the sample shall be rejected.

4.4.5 Particle size. The particle size test shall be done according to ASTM D 502 using a No. 20 and a No. 100 sieve.

4.4.6 pH. The pH of a 1.0 percent solution shall be determined according to ASTM E 70.

4.4.7 Interfacial tension. The interfacial tension of a 1.0 percent solution shall be determined by using ASTM D 1331A in triplicate.

4.4.8 Polyphosphates. Make a 1.0 percent solution of the detergent in distilled water. Pipet a 4 ml aliquot into a 1000 ml volumetric flask and fill to volume with distilled water. This will be a one ppm solution of phosphate species from a detergent sample containing 25 percent phosphate. Determine the hydrolyzable phosphorous content (polyphosphates) in accordance with ASTM D 515.

4.4.9 Rinsibility. Place 5 gms of the material on a clean glass plate about four (4) inches by six (6) inches by 0.08 inches. Add sufficient water to make a thin paste. Spread this paste evenly on one side of the piece of glass. Then rinse the glass plate by shaking it for three 10-second periods in successive 2-liter beakers of distilled water at 190°F (88°C). Inspect the plate visually for any adherent cleaner. Hold the plate vertically for 45 seconds, then inspect for the presence of water droplets on the surface.

4.4.10 Solubility. One gram of the product shall be completely soluble in 100 ml of synthetic hard water at 25°C within 10 minutes. The hard water used in this test shall be made according to the following formula: 0.264 gm/liter $CaCl_2.2H_2O$ plus 0.295 gm/liter $MgSO_4.7H_2O$.

4.4.11 Water softening. Dissolve 2.27 gm $CaCl_2.2H_2O$ and 2.52 gm $MgSO_4.7H_2O$ in separate 100 ml volumetric flasks and fill to volume with distilled water. Pipet a 10 ml aliquot of each of these solutions into one 500 ml volumetric flask and fill to volume with distilled water. Place 2.5 grams of the detergent sample into a 500 ml volumetric flask, and fill to volume with distilled water. Place 50 ml of each solution into a beaker. Transfer an additional 50 ml aliquot from the 500 ml flask containing the detergent sample solution to a 100 ml volumetric flask and fill to the mark with distilled water. This is the reference solution. Pipette 10 ml of the solution from the beaker into a 20 mm by 150 mm test tube. Pipette 10 ml of the reference solution into an identical test tube. Heat the test tubes in a water bath at 66°C for 5 minutes and allow to cool for 30 minutes. There shall be no flock or precipitate formed in the hard water sample.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. Five pounds of detergent shall be packaged in a metal can conforming to PPP-C-96, type V, class I; type V, class 2; or type VI; in a plastic jar with a recloseable opening; or in a water resistant chipboard carton with a recloseable opening.

5.1.2 Level C. The detergent shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Six containers of detergent shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed, waterproofed, and banded in accordance with the appendix to the box specification. For the plastic jars, the box shall be fitted with a full height fiberboard divider or separator which is 3/8 inch ± 1/8 inch higher than the jars. The fiberboard shall be the same class and grade as the box. The cartons and cans do not require these dividers.

5.2.2 Level B. Six containers of detergent shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the specification. For the plastic jars, the box shall be fitted with a full height fiberboard divider or separator which is 3/8 inch ± 1/8 inch higher than the jars. The fiberboard shall be the same class and grade as the box. The cartons and cans do not require these dividers.

5.2.3 Level C. Detergent packaged as specified in 5.1 shall be packed in containers of sufficient quality to assure carrier acceptance and safe arrival at destination. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. Marking of the interior and shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies and MIL-STD-129 for military agencies, as applicable (see 6.2).

6. NOTES

6.1 Intended use. The detergents covered in this specification are intended for use in cleaning laboratory glassware and stainless steel articles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing (see 5.1 and 5.2).
(c) Applicable marking document (see 5.3).

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
HEW - FEC
HEW - HSM
DMS - VA
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PREPARING: GSA - FSS

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